

ENGLISH

SUPER-FOAM

Low-Pressure Micro Physical Foaming Series



WOOJIN
PLAIMM

Improve Quality Up Point

- Experience **Enhanced** Quality

1 Quality improvement

- | Reduction of shrinkage and deformation
 - ① Improved quality of injection molded products
 - ② Elimination of sink marks
 - ③ Increased dimensional accuracy and reduced warpage

2 Versatility

- | A certain amount of physical foam molding is possible regardless of type without using a foam-only resin.



3 Increased productivity

- | Increase productivity by shortening the overall process time as the suppression process is omitted.

4 Stability & Other Effects

- | Enables continuous stable foaming injection at the level of general injection.
- | Insulation, Insulation, Insulation, Buffering, Floating (floating), Sound Insulation, Absorption.

5 Durability & Precision Position Control

- | Because of the low pressure GAS, the load on the machine is low and the maintenance cost is low.
- | The A5 Series fully implements core-back function with precision position control technology, which makes foam molding effective.

Saved Quality Down Point

- Experience **Greater Efficiency** and Quality

1 Light weighting

- | Reduction of raw material leads to light weightiness of the product.
- | Super-foam provides the solution to automotive market and market where aims to light weightiness.

2 Cost reduction

- | Foam molding can reduce the amount of raw materials that needed so it can produce the product with small amount of raw material.

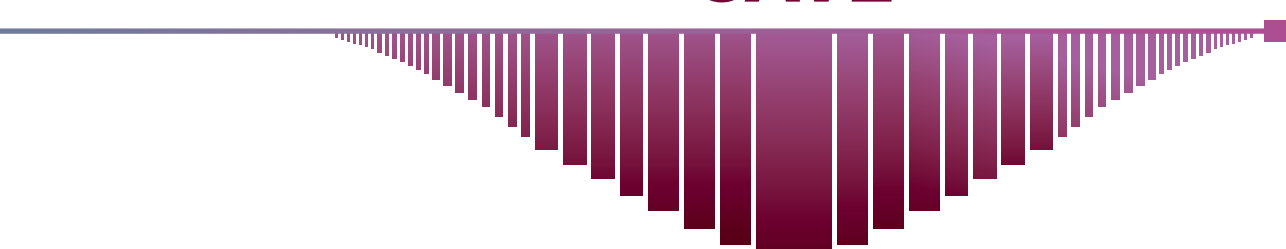
3 Energy reduction

- | Foam molding omitted the holding process.
- | As a result, only about 60 to 70 percent of the clamping force is required compare to normal injection.

4 Other Savings

- | Various firing items can be applied in advance through the in-house showroom.
- | Reduce development time and reduce post-process and assembly time by reducing shrinkage deformation.

SAVE

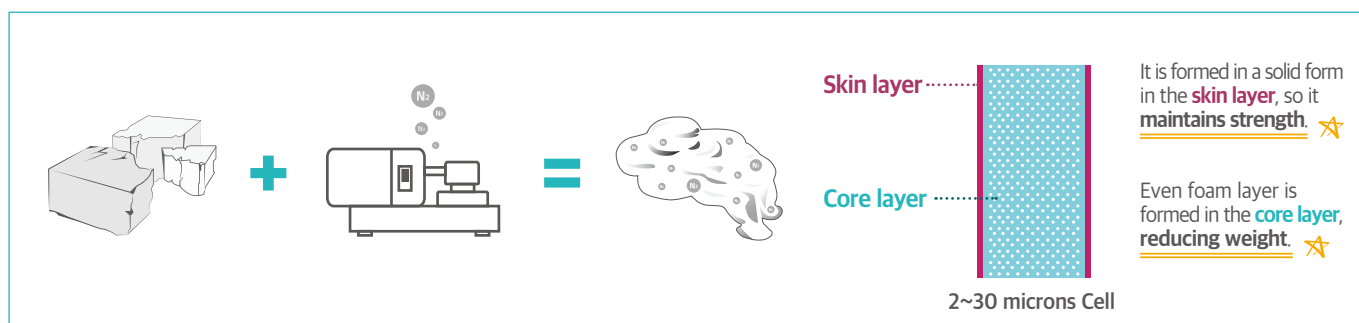


5 Equipped with Metering Motor

- | It realizes simultaneous movement during charging by equipping electric charging motor and energy saving due to high power transfer efficiency.

SUPER-FOAM

SUPER-FOAM TECHNOLOGY

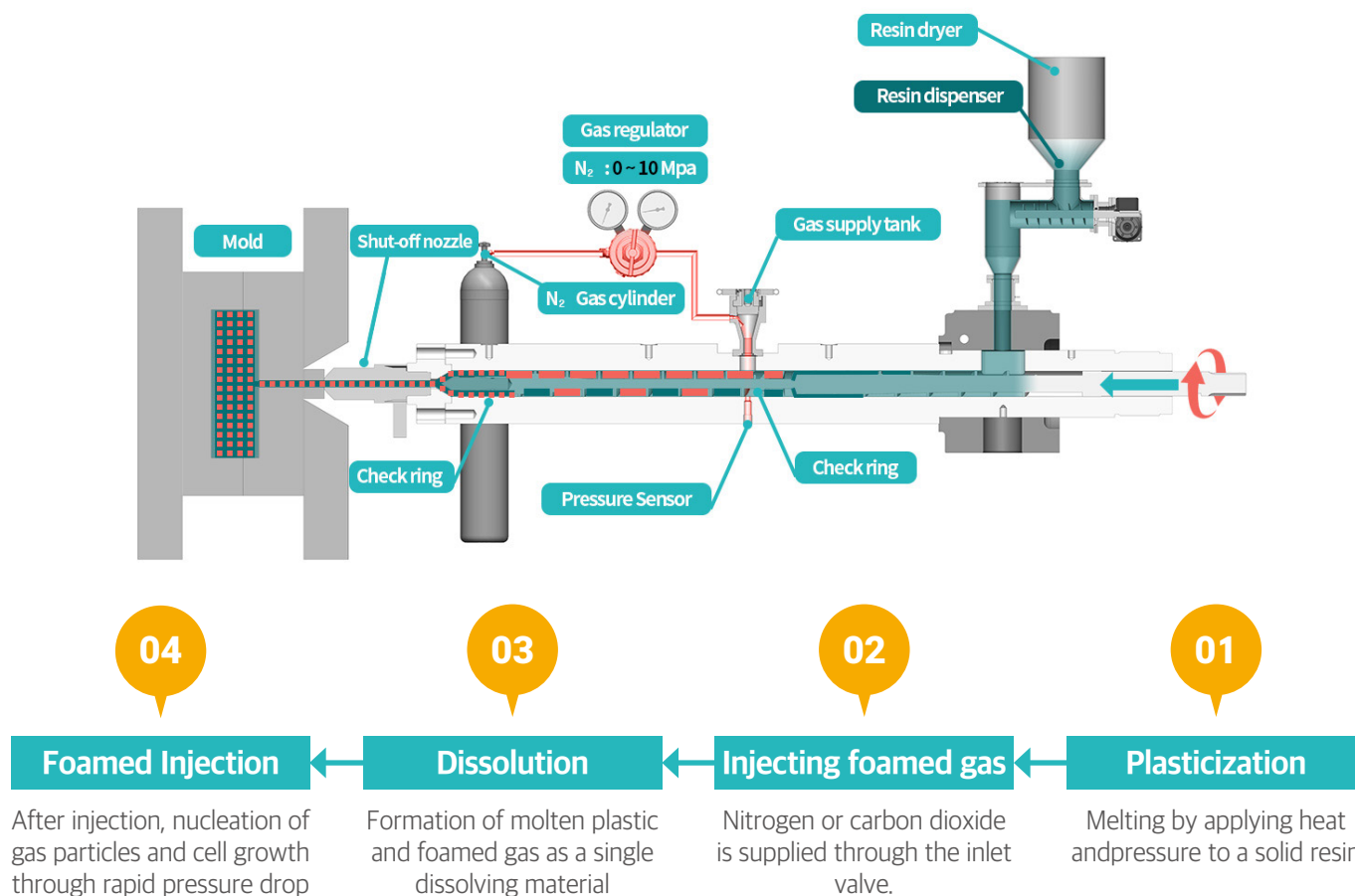


Super-Foam is a technology that disperses and dissolves nitrogen in a molten polymer to generate bubble nuclei, which are decompressed during injection, and grows and stabilizes them to form porous products.

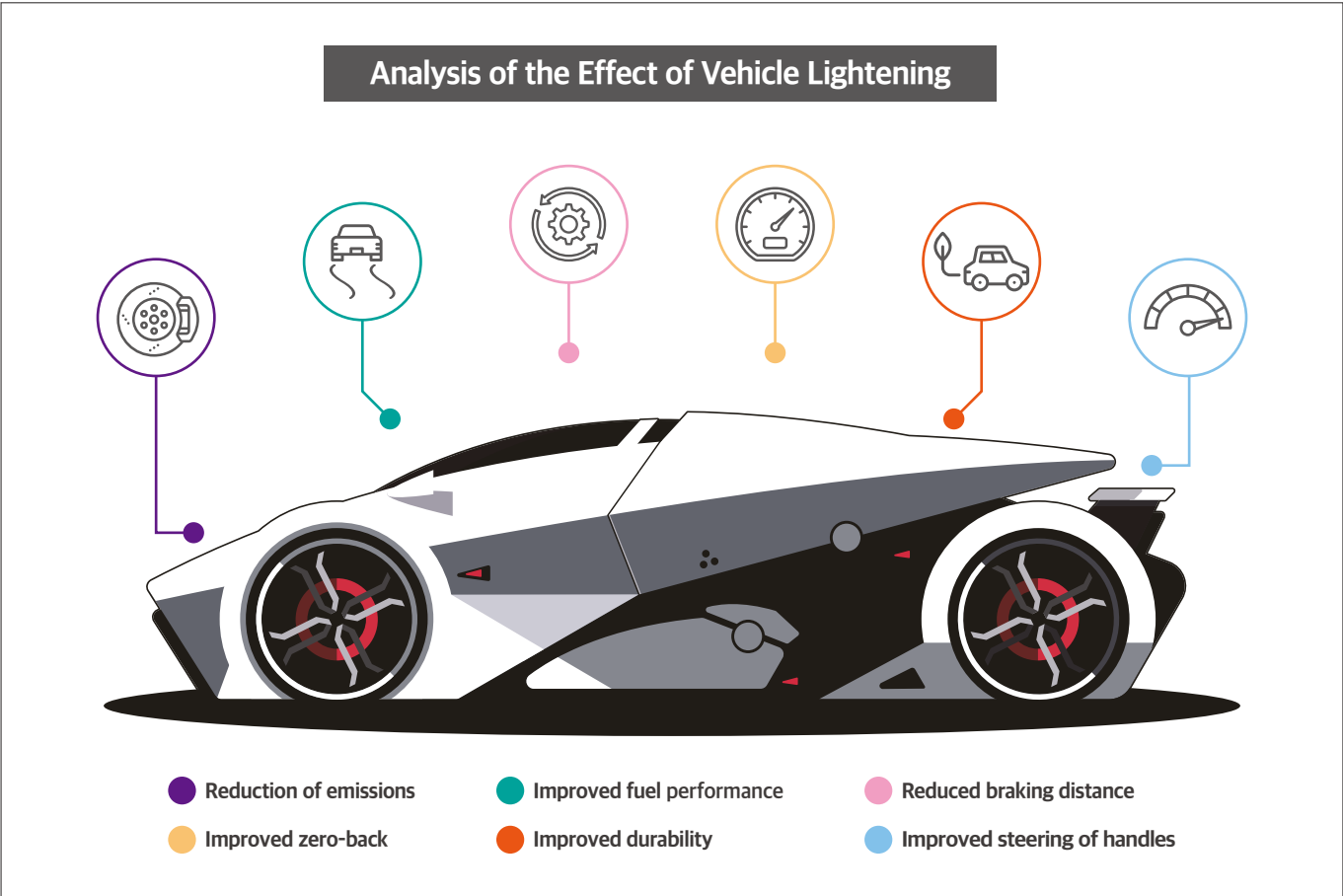
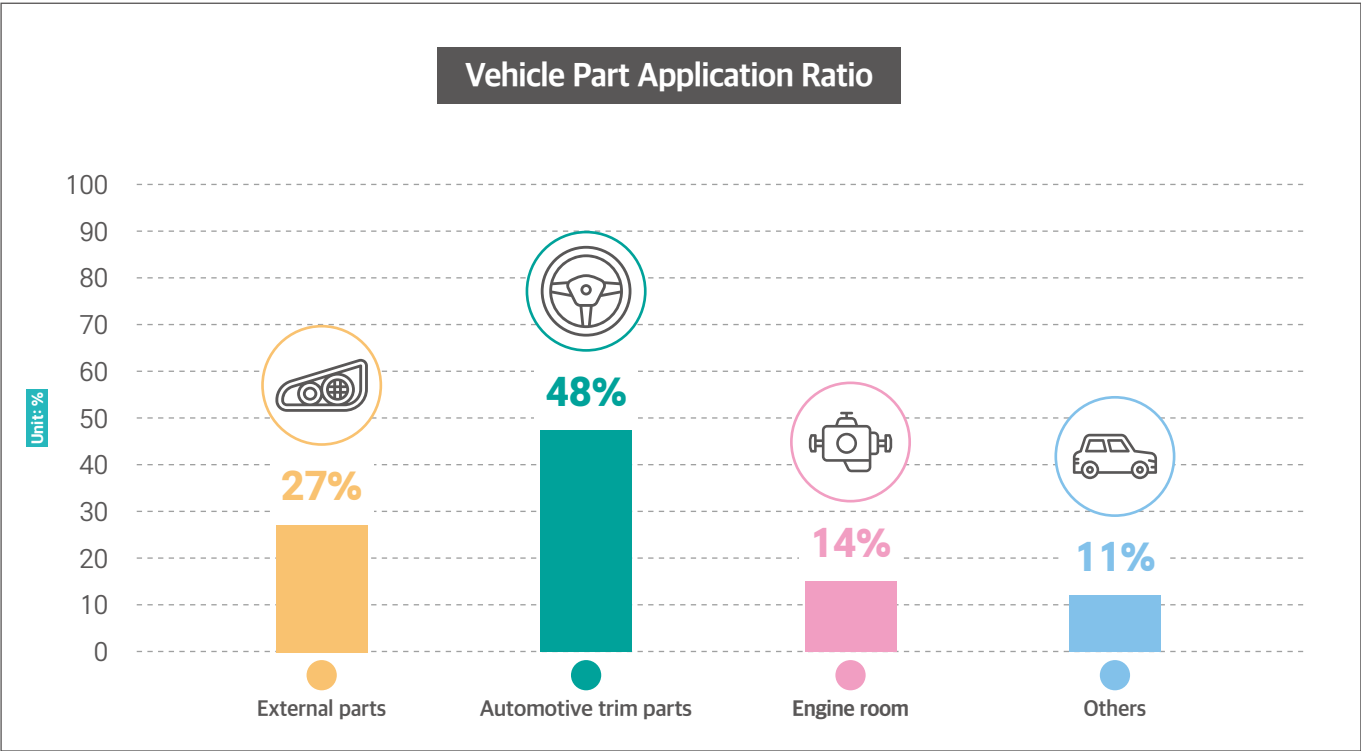
Category	Amount	Maximum injection pressure	Holding switchover pressure	Holding pressure time	Holding pressure	Barrel set temperature	Clamping force
Solid	32mm	650kg/cm ²	622kg/cm ²	4sec	650kg/cm ²	220~180	1736kn
15% foamed	26.7mm	557kg/cm ²	233kg/cm ²	-	-	205~165	616kn
Reduce effect	↓16.6%	↓14.4%	↓62.5%	↓4sec	↓650kg/cm ²	↓15	↓65.6%

※ Foaming ratio can be applied up to 80% depending on the product, resin, and mold characteristics

SUPER-FOAM PROCESS



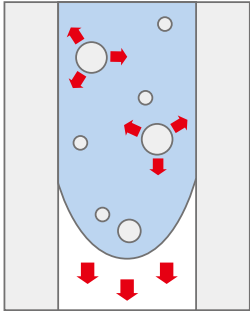
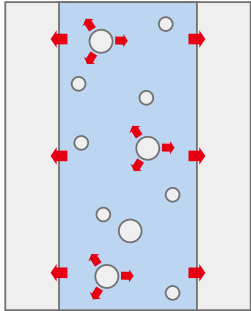
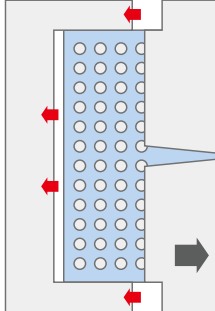
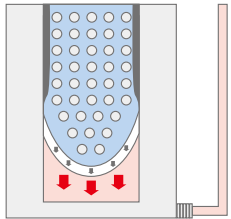
BACKGROUND



SUPER-FOAM

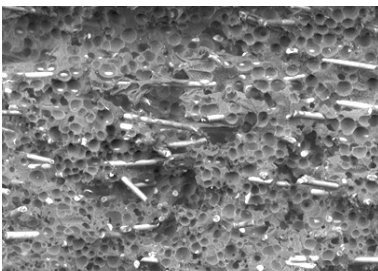
TYPES OF SUPER-FOAM

(Foaming ratio may vary depending on mold and resin conditions)

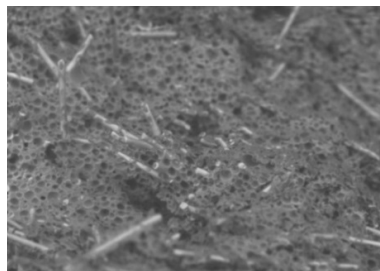
Method	Short-Shot	Full-Shot	Core-back	CPM
Types				
Foaming rate	10~20%	3~5%	20~60%	20~70%
Effects	Light weight, lower material costs	Quality improvement (Export, Deformation Prevention)	Reduce material costs, reduce weight	Reduce material costs, reduce weight, and surface improvement

SUPER-FOAM WITH VARIOUS TYPES OF RESINS

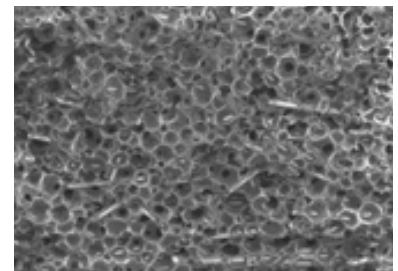
Magnified View of Foamed Injection Molding by Resin Type



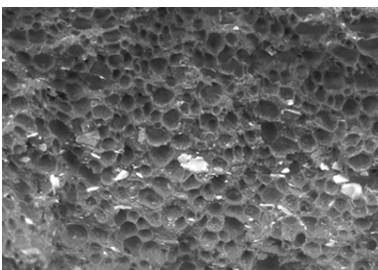
PA6 + GF 30%



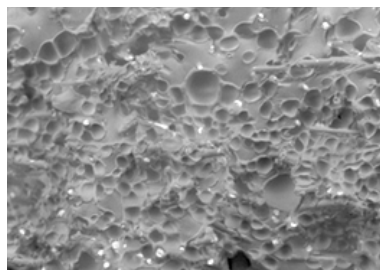
PBT + GF 30%



PA66 + GF 33%



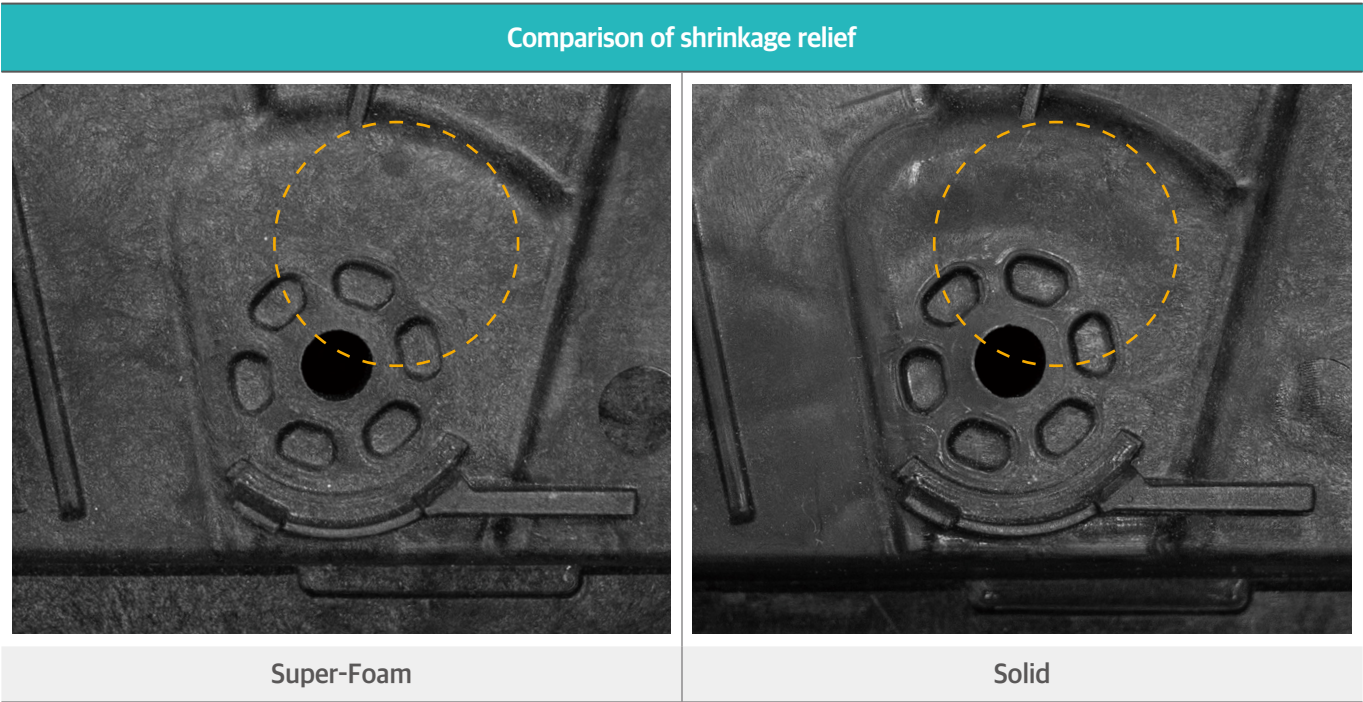
PP + Talc 16%



PC + GF 20%

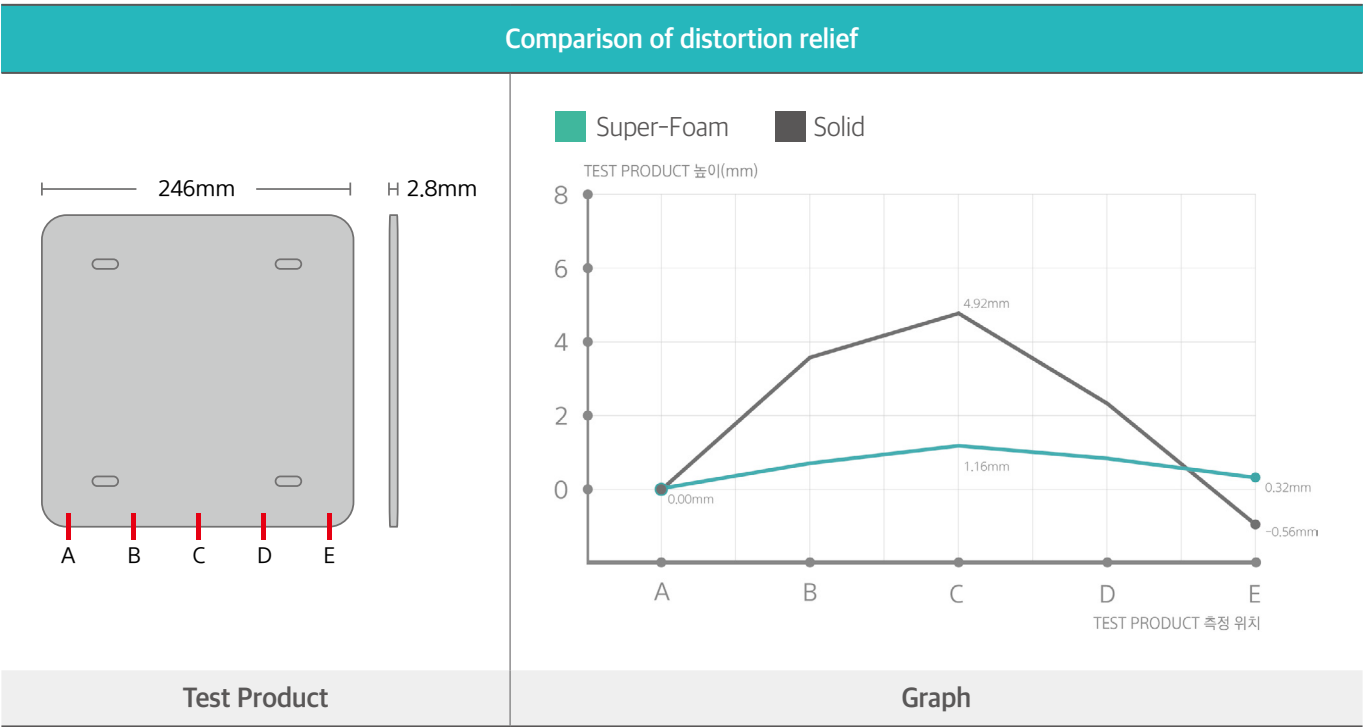
Foaming Criteria : N2 pressure 60bar

COMPARISON OF SHRINKAGE RELIEF



COMPARISON OF DISTORTION RELIEF

Bending Comparison between Solid Injection and Foam Injection



Solid (Standard Deviation)=2.36
Super-Foam (Standard Deviation)=0.48

CLEAN-FOAM

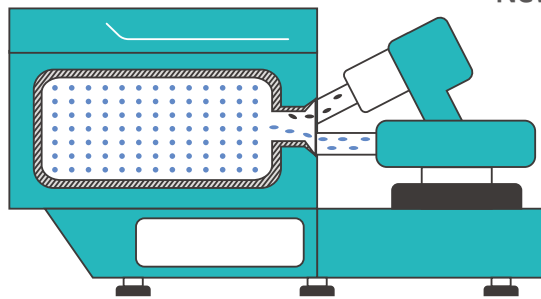
CLEAN-FOAM TECHNOLOGY

The “**Clean-Foam**” method is an eco-friendly foam injection method that injects a clean surface without sinks. The skin layer is molded with new materials, and the core layer is foam-molded using new or recycled materials using WOJIN PLAImm’s unique “Super-foam” method.

※ Clean-Foam is an eco-friendly, next-generation injection molding technology

Clean-Foam technology allows for 100% suppression of the surface foaming effect of conventional foaming.

- ① Effect: Produces a smooth, fine-grained surface with consistent quality without surface foaming (irregularity).
- ② Usage: It can be painted and used in various ways, such as exterior material.



New / Renewable Materials :  ... Smooth surface treatment
(100% suppression of surface foaming)

Foam :  ... Energy saving, cost reduction

Clean-Foam Merit

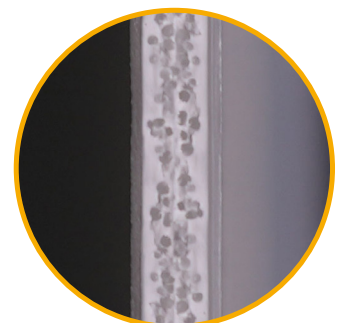
- ① Production cost reduction / Product weight reduction
- ② 100% improvement in surface foaming problems
- ③ Sink-free clean surfaces (immediately available for painting)



General foam

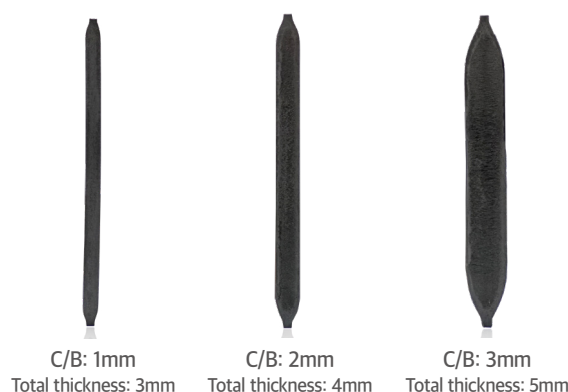
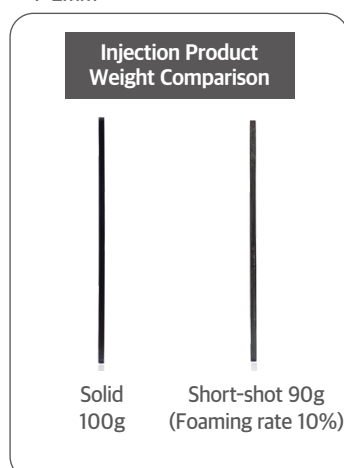


WOJIN PLAImm Clean-Foam



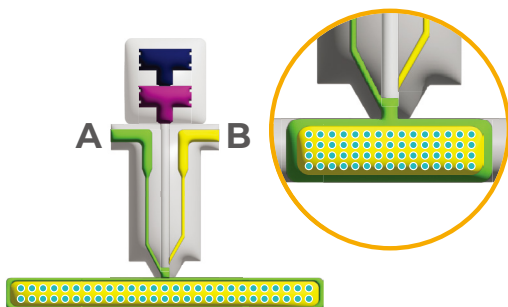
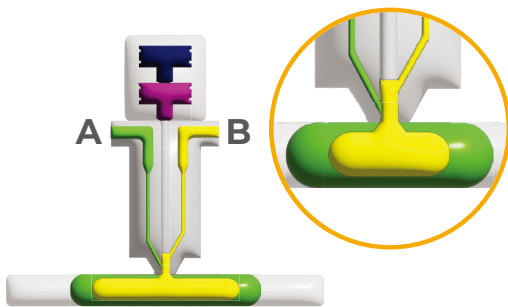
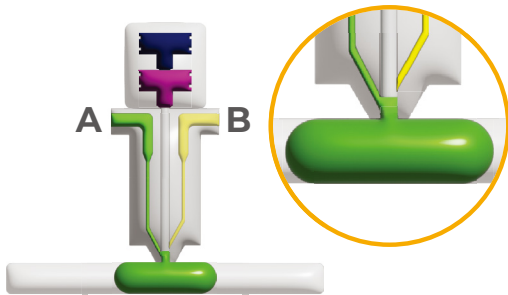
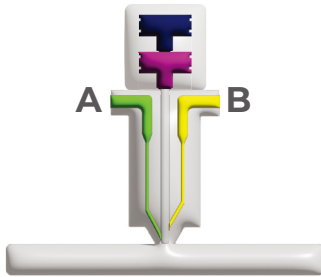
Clean-Foam
Cross-Sectional Shape

T: 2mm



Low-pressure micro-physical foaming has a **20%** material cost reduction effect compared to solid injection molding.

CLEAN-FOAM INJECTION PROCESS



STEP 1

Stand By

| Controls A or B channels using a hot runner system.

STEP 2

Skin Injection

| The new material is injected through the A channel to form a skin layer.

STEP 3

Core Injection

| New or recycled materials are foamed through the B channel. The foamed resin forms a core layer.

STEP 4

Skin Injection

| The new material is injected into the A channel to suppress surface foaming and sinking. The "Clean-Foam" technology, a foam injection molding technique with a clean surface, is completed.

OVERVIEW

IDL-AS SUPER-FOAM

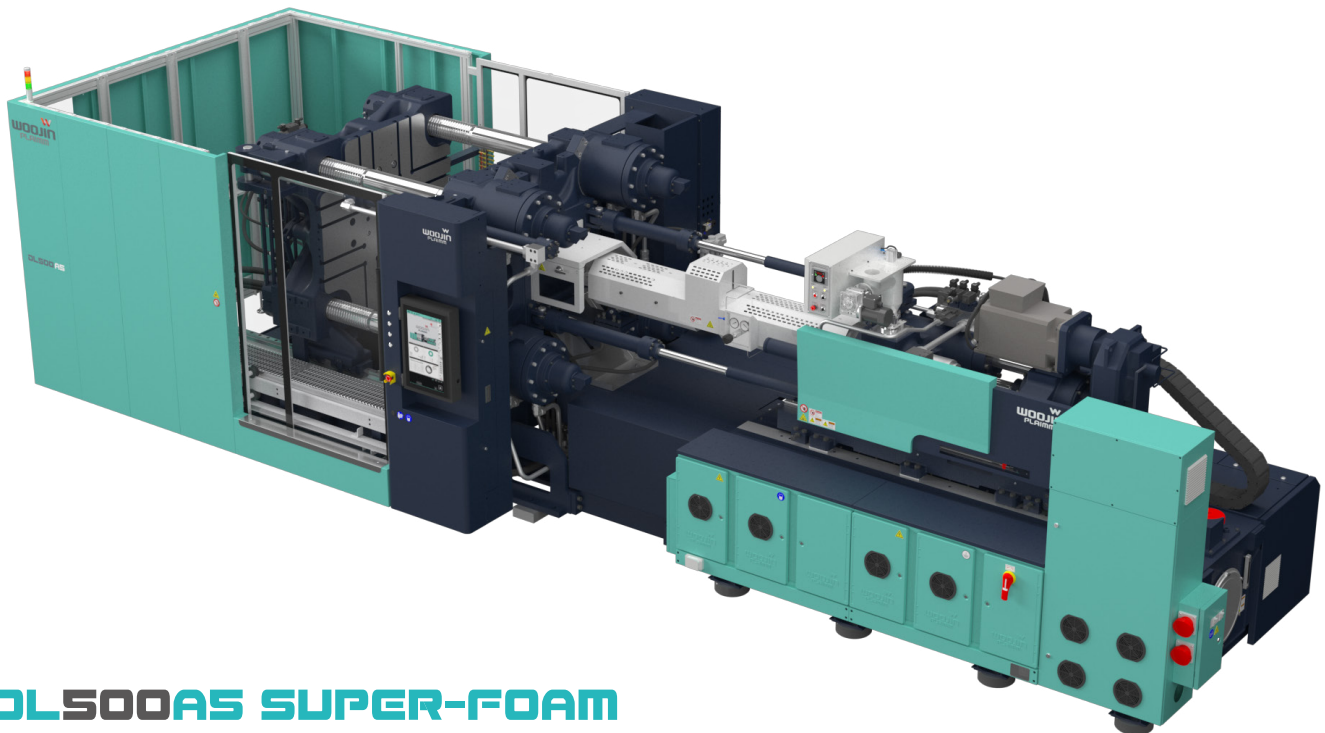
Premium Energy-Saving Two-Platen Direct Clamping LFIM Series

This is a foam injection mold that dissolves inert GAS under the threshold (less than 10 Mpa) into the molten resin to inject it into the mold, and utilizes the fine discharge of GAS.

(500~3300Ton)



IMM Promotion



DL500AS SUPER-FOAM

※ Product images may be subject to change without prior notice due to optional specifications or product improvements.



1 21 inch Full HD Controller

- | Intuitive and visual custom interface.
- | Multi Touch Screen Controller for optimal usability.



2 Quantitative Supply System

- | Reduce screw and barrel by supplying 70-90% raw material within the screw.
- | A large amount of cells are formed by securing the volume of gas through quantitative supply system.

FEATURES

DL-A5(S.F.) CLAMPING UNIT

1 Platen Structure

| Higher mold installation weight by minimizing platen bending through FEA design (ver.2)

3 Increased Durability

| The application of U-shaped screws and tie bars increases durability due to stress dispersion effects.

5 Precise Positioning

| Positioning improved by mounting a high-responsive proportional valve.

2 Maximized Space Utilization

| Maximize the space utilization by easy installation and detachment of mold in low and narrow spaces (ver.2).

4 Cycle Time Reduction

| Synchronized Half-nut locking controlling system to shorten the cycle time.

6 Stable Mold Opening and Closing

| Stable mold opening and closing by symmetrical booster cylinders.

DL-A5(S.F.) INJECTION UNIT

1 Injection Back Pressure Closed-loop Control

| Keep constant back pressure set by user through closed-loop control.

3 Reduced Maintenance Cost

| Equipped with a gas injection system that allows IN-HOUSE maintenance

| Metered supply system reduces wear on the screw and cylinder

2 Application of Electric Metering Motor

| Precise metering ensures excellent repeatability and enables simultaneous multi-actions, improving productivity while reducing noise and energy consumption

4 Energy Savings

| Application of L/M guides to the injection bed and metering section maintains parallelism and reduces friction, resulting in energy savings

DL-A5(S.F.) HYDRAULIC UNIT

1 Special Coating for Oil Tank

| Improve internal cleanliness through special coating inside and outside of the hydraulic block.

| Prevent corrosion due to moisture and oil evaporation by specially coated oil tank.

3 Leakage Prevention in Hydraulic Pipes

| Seamless pipes, expansion tubes, and EO2-FORM piping connections robust against high pressure and vibration prevent leakage and damage on parts.

2 Independent Hydraulic Oil Circulation System

| Keep cleanliness and constant temperature through independent oil tank circulation system.

(contribute to longer lifespan of hydraulic oil and improved durability of hydraulic components)

4 Reduction of Energy and Cooling Water Consumption

| Servo pump system minimize idle operation so as to reduce energy and cooling water consumption.

OVERVIEW

ITE-AS SUPER-FOAM

Ultra Energy-Saving Toggle Electric LFIM Series



IMM Promotion

This is a foam injection mold that dissolves inert GAS under the threshold (less than 10 Mpa) into the molten resin to inject it into the mold, and utilizes the fine discharge of GAS.

(110~850Ton)



TE220AS SUPER-FOAM

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1 21 inch Full HD Controller

- | Intuitive and visual custom interface.
- | Multi Touch Screen Controller for optimal usability.



2 Quantitative supply system

- | Reduce screw and barrel by supplying 70-90% raw material within the screw.
- | A large amount of cells are formed by securing the volume of gas through quantitative supply system.

FEATURES

TE-A5(S.F.) CLAMPING UNIT

1 New Toggle Applied

| Application of new toggle system for fast mold opening and closing and clamp-type toggle pin fixing to increase the durability.

3 High Energy Efficiency

| High energy efficiency due to the high speed mold opening and closing, maintaining constant platen parallelism and little friction.

5 Wide Tie-bar Distance

| Applicable with a wide range of molds by increased tie-bar distance and mold opening compared with equivalent group of other brands.

2 Automatic Clamping Force Adjustment

| Automatic clamping force adjustment by detecting real-time clamping force variation.

4 Center Press Method

| Center press method delivers excellent clamping power to mold and uniform surface pressure.

TE-A5(S.F.) INJECTION UNIT

1 Injection Back Pressure Closed-loop Control

| Keep constant back pressure set by user through closed-loop control.

3 Reduced Maintenance Cost

| Equipped with a gas injection system that allows IN-HOUSE maintenance
| Metered supply system reduces wear on the screw and cylinder

2 Application of Electric Metering Motor

| Precise metering ensures excellent repeatability and enables simultaneous multi-actions, improving productivity while reducing noise and energy consumption

4 Energy Savings

| Application of L/M guides to the injection bed and metering section maintains parallelism and reduces friction, resulting in energy savings



OVERVIEW

ISFV

Vertical Low Pressure Micro-Physical Foaming Hybrid Series



IMM Promotion

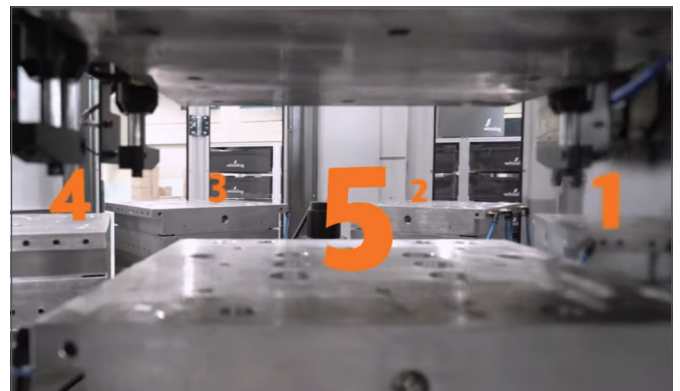
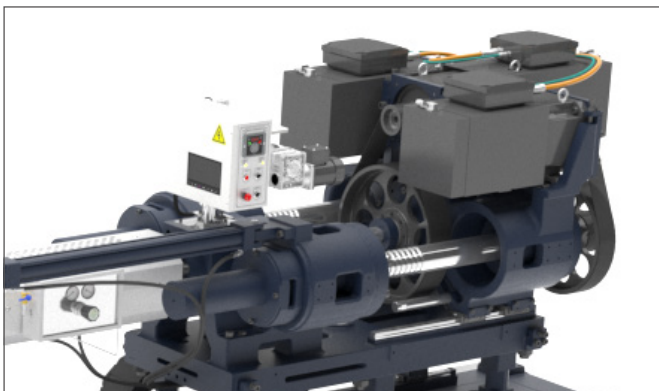
A low-pressure micro-physical foaming injection molding machine that allows multiple mold installations and enables fast processing

(Ø1800 | Ø2400)



SFV2400

※ Product images may be subject to change without prior notice due to optional specifications or product improvements.



1 Physical Foam

- | Resin and nitrogen gas dissolved in the barrel undergo rapid pressure drop during injection, inducing nucleation of gas particles and cell growth to form the product
- | Effects: Product lightweighting, dimensional stability, and reduced warpage

2 Reduce Cycle Time

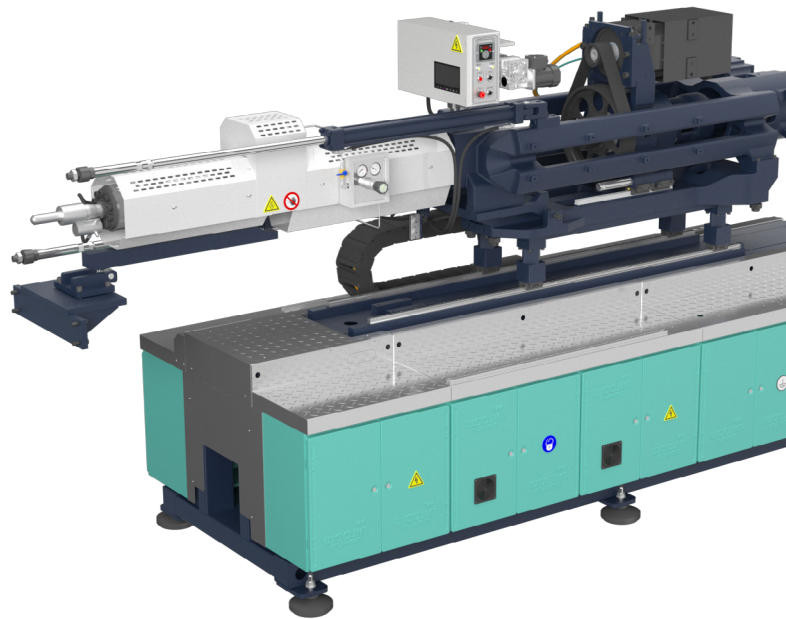
- | The turntable with 5 molds reduces cycle time for products with long cooling times and can handle 5 different shapes and sizes.
- | Minimization of holding pressure time.

FEATURES

SFV INJECTION UNIT

1 Integrated Injection Structure

- | High-rigidity integrated housing unit.
- | Stability during injection with dual full nozzle touch cylinder.
- | Fast injection response and improved positioning precision with 0.01mm control.
- | Metering drive LM guide and friction resistance optimization for easy precision injection with back pressure close
- | Electric injection system reduces utility costs



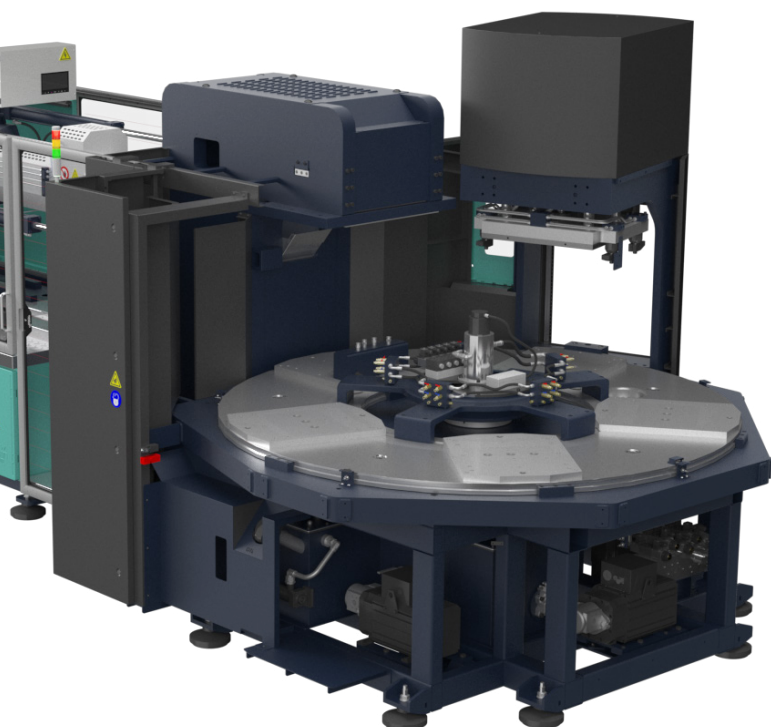
SFV CLAMPING UNIT

1 C-Frame Clamping Structure

- | Tie-barless structure improves mold accessibility and facilitates installation of automation equipment
- | Easy installation of peripheral equipment such as hydraulic cores, cooling, and air lines

2 Low-Profile Rotary Table

- | Low turntable height improves the working environment and allows easy mold mounting/dismounting, reducing changeover time
- | Hydraulic system is located at the lower frame for easy maintenance
- | Fast-response servo control of the turntable motor enhances positioning accuracy and reduces 180° rotation time
(Positioning accuracy: ± 0.01 mm / Rotation speed: 1.8 sec)

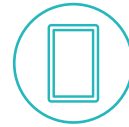


IMC 510



* The Red Dot Award is a globally prestigious design award, and together with Germany's iF Design Award and the USA's IDEA, it is recognized as one of the world's top three design awards

FEATURES



- Equipped with B&R Controller
- 21-inch Full HD Touch Screen
- User Interface (UI/UX) Optimized for IMM



- Injection process design using Smart Sequence
- Review processes and receive feedback through PLAIMM-X
- Process management analysis and energy monitoring

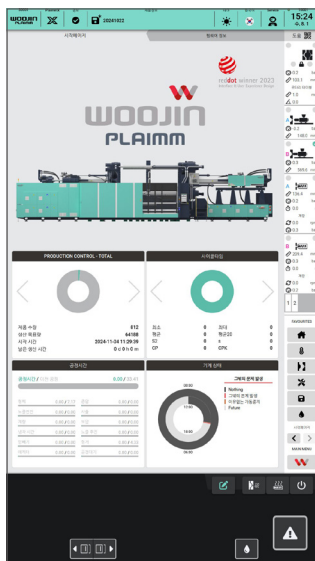


- Supports OPC-UA (EUROMAP 77)
- Data provided via SQL and CSV
- Peripheral device monitoring and remote support services available

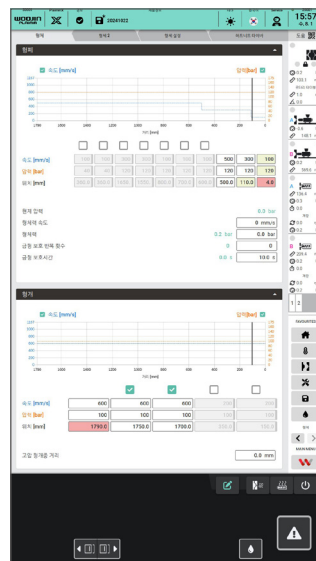


- Check operation sequence with Overview Chain
- View clamping and injection graphs on a single page
- Frequently used functions managed through custom tabs

MAIN PAGE SCREEN CONFIGURATION



Home



Mold



Injection/Pressure Hold



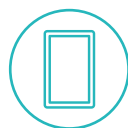
Metering

※ This controller may be changed without prior notice due to optional specifications or software updates.

IMC 710



FEATURES

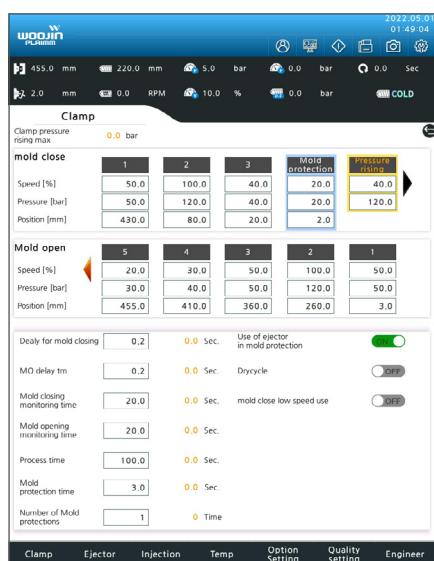


- UI icons in image-centered pictogram style
- Interface optimized with essential buttons
- Supports multiple languages

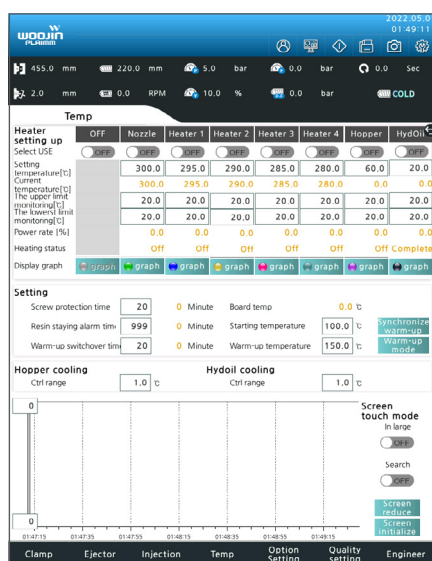
SPECIFICATIONS

Display	15-inch (IPS LCD)
Resolution	768 X 1024
Touch Screen	Capacitive
CPU	ARM Cortex-A9 Core 11,500 DMIPS
RAM	DDR3, 1GB
Storage	eMMC 8GB
IP Rating	IP65 (Front) IP20 (Back)

MAIN PAGE SCREEN CONFIGURATION



Clamp



Temperature



Machine Information

※ This controller may be changed without prior notice due to optional specifications or software updates.

SPECIFICATIONS

DL-A5 SUPER-FOAM

MODEL		DL500A5	DL600A5	DL700A5	DL900A5	DL1100A5	DL1300A5
Injection Unit		IH2800	IH4200	IH5900	IH8800	IH8800	IH11900
Screw & Barrel		A	A	A	A	A	A
Screw diameter	mm	70	80	90	105	105	115
Injection pressure	kg/cm ²	1889	1887	1885	1756	1756	1809
	Mpa	185	185	185	172	172	177
Theoretical injection volume	cm ³	1482	2212	3149	5022	5022	6544
Shot weight(PS)	g	1365	2038	2902	4628	4628	6030
Injection rate	cm ³ /s	472	602	763	1041	1041	1249
Screw stroke	mm	385	440	495	580	580	630
Injection speed	mm/s	123	120	120	120	120	120
Plasticizing capacity (PS)	kg/h	252	328	408	515	515	607
Screw rotation speed	rpm	180	165	150	125	125	115
Clamping Unit							
Clamping force	ton(kN)	500(4903)	600(5884)	700(6865)	900(8826)	1100(10787)	1300(12749)
Mold opening force	ton(kN)	38(368)	45(441)	53(515)	68(662)	83(809)	98(956)
Distance between tie-bar(HxV)	mm	920 x 830	1040 x 910	1110 x 1110	1200 x 1120	1420 x 1170	1580 x 1280
Platen dimension (HxV)	mm	1280 x 1260	1430 x 1370	1520 x 1490	1720 x 1610	1870 x 1820	2230 x 1990
Daylight	mm	1650	1750	1850	2100	2400	3050
Min. mold height	mm	350	400	450	500	600	700
Max. mold height	mm	900	950	950	1100	1200	1400
Ejector force	ton(kN)	11.1(108.9)	16.6(162.8)	19.8(194.2)	26.9(263.8)	26.9(263.8)	34.4(337.3)
Ejector stroke	mm	200	220	250	250	250	300
Dry cycle time	sec	3.3	3.3	3.3	4.0	4.4	5.0
Max. mold weight (Fixed / Moving / Total)	ton	5.3 / 5.3 / 8	6.7 / 6.7 / 10	7.3 / 7.3 / 11	8.6 / 8.6 / 13	14 / 14 / 21	20 / 20 / 30
General							
Heater capacity	kW	28.7	37.3	47.0	62.4	62.4	76.4
Motor capacity	kW	65.2	87.6	87.6	110	110	142.6
Total electric power capacity	kW	93.9	124.9	134.6	172.4	172.4	219.0
Hydraulic oil tank capacity	L	600	800	800	920	920	1150
Machine weight (Clamping+Injection)	ton	19(13.5+5.5)	26(17+9)	32(21.5+10.5)	41(29+12)	50(37.5+12.5)	72(55+17)
Machine dimension (LxWxH)	m	8.4 x 2.7 x 2.2	8.6 x 2.9 x 2.2	9.3 x 3.1 x 2.4	10.5 x 3.4 x 2.5	10.7 x 3.4 x 2.7	12.3 x 4.2 x 3.4
Cooling water consumption	L/min	130	130	130	180	180	180

01. The oretical injection volume: cross section of screw*screw stroke.
 02. The minimum mold size must be at least 70% of the tie bar distance.
 03. The specifications might be changed without any prior notice.

04. The metering distance is recommended in the arnge of 1 to 3 times of screw diamteer.
 05. Machine dimensions may vary when the injection unit moves UP or DOWN.

DL1800A5	DL2000A5	DL2300A5	DL2500A5	DL2700A5	DL3000A5	DL3300A5
IH15300	IH15300	IH15300	IH21500	IH21500	IH33000	IH33000
A	A	A	A	A	A	A
125	125	125	140	140	160	160
1743	1743	1743	1816	1816	1800	1800
171	171	171	178	178	177	177
8406	8406	8406	11853	11853	16085	16085
7746	7746	7746	10923	10923	14822	14822
1349	1349	1349	1537	1537	1719	1719
685	685	685	770	770	800	800
110	110	110	100	100	85	85
692	692	692	850	850	1000	1000
105	105	105	95	95	78	78
1800(17652)	2000(19613)	2300(22555)	2500(24517)	2700(26478)	3000(29420)	3300(32326)
135(1324)	150(1471)	173(1692)	188(1839)	203(1986)	225(2206)	248(2427)
1850 x 1610	2020 x 1610	2020 x 1610	2180 x 1760	2180 x 1760	2260 x 1810	2260 x 1810
2450 x 2200	2600 x 2250	2600 x 2250	3030 x 2610	3030 x 2610	3140 x 2660	3140 x 2660
3400	3600	3600	3900	3900	4000	4000
700	800	800	900	900	1100	1100
1600	1700	1700	2000	2000	2000	2000
44.5(436.4)	44.5(436.4)	44.5(436.4)	67.8(664.9)	67.8(664.9)	67.8(664.9)	67.8(664.9)
300	300	300	350	350	350	350
5.8	5.8	5.8	8.2	8.2	8.2	8.2
30 / 30 / 45	41 / 41 / 62	41 / 41 / 62	50 / 50 / 75	50 / 50 / 75	56 / 56 / 85	56 / 56 / 85
86.2	86.2	86.2	110.0	110.0	209.0	209.0
142.6	142.6	142.6	165.0	165.0	220.0	220.0
228.8	228.8	228.8	275.0	275.0	429.0	429.0
1450	1450	1450	1650	1650	2650	2650
115(96+19)	115(96+19)	115(96+19)	143(121+22)	143(121+22)	180(149+31)	180(149+31)
13.7 x 4.2 x 3.4	14.1 x 4.5 x 3.4	14.1 x 4.5 x 3.4	16 x 4.7 x 3.7	16 x 4.7 x 3.7	17.8 x 5 x 4	17.8 x 5 x 4
180	180	180	240	240	240	240

SPECIFICATIONS

TE-A5 SUPER-FOAM

MODEL		TE110A5	TE170A5		TE220A5		TE280A5	TE280WA5
Injection Unit		IE 260	IE 370		IE 520		IE 720	IE 1000
Screw & Barrel type		A	A	B	A	B	B	B
Screw diameter	mm	32	36	40	40	45	50	55
Injection pressure	kg/cm²	2024	2033	1647	2059	1627	1667	1690
	Mpa	198	199	162	202	160	163	166
Injection holding pressure	kg/cm²	1822	1830	1482	1853	1464	1500	1521
	Mpa	179	179	145	182	144	147	149
Theoretical injection volume	cm³	129	183	226	251	318	432	594
Shot weight (PS)	g	117	167	206	228	289	393	541
Injection rate (standard)	cm³/s	173	204	251	201	254	295	356
Injection rate (optional)	cm³/s	346	407	503	402	509	589	713
Screw stroke	mm	160	180	180	200	200	220	250
Injection speed (standard)	mm/s	215	200	200	160	160	150	150
Injection speed (optional)	mm/s	430	400	400	320	320	300	300
Plasticizing capacity (PS)	kg/h	64	86	117	117	158	213	273
Screw rotation speed	rpm	400	375	375	375	375	375	300
Clamping Unit								
Clamping force	ton(kN)	110(1078)	170(1667)		220(2157)		280(2745)	280(2745)
Distance between tie-bar (H×V)	mm	470x470	570x570		625x625		670x670	720x720
Platen dimension (H×V)	mm	680x680	840x790		900x870		970x980	1020x1020
Daylight	mm	400	500		550		600	650
Max. Daylight	mm	850	1000		1150		1250	1350
Min. mold height	mm	150	180		200		250	300
Max. mold height	mm	450	500		600		650	700
Ejector force	ton(kN)	3.1	3.4		3.4		4.3	4.3
Ejector stroke	mm	120	150		180		200	200
General								
Heater capacity	kW	9.7	12.1	13.7	13.7	15.3	20.6	22.3
Machine weight	ton	5.5	7.5		10.5		14.5	17.5
Machine dimension (L×W×H)	m	5.8×1.2×2.0	6.0×1.6×2.0		6.5×1.7×2.1		7.2×1.7×2.0	7.7×1.8×2.1

01. The oretical injection volume: cross section of screw*screw stroke.
 02. The minimum mold size must be at least 70% of the tie bar distance.
 03. The specifications might be changed without any prior notice.

04. The metering distance is recommended in the arnge of 1 to 3 times of screw diamteer.
 05. Machine dimensions may vary when the injection unit moves UP or DOWN.

TE350A5	TE400A5	TE450A5	TE550A5	TE650A5	TE850A5
IE 1360	IE 1700	IE 2800	IE 4000	IE 5700	IE 8000
B	B	A	A	A	A
60	65	70	80	90	105
1716	1712	2083	2034	2009	1734
168	168	204	199	197	170
1544	1541	1875	1831	1808	1561
151	151	184	180	177	153
763	995	1347	2011	2863	4286
694	905	1226	1830	2605	3900
424	498	577	754	954	1299
848	995	1155	-	-	-
270	300	350	400	450	495
150	150	150	150	150	150
300	300	-	-	-	-
233	259	279	347	408	515
250	225	200	175	150	125
350(3432)	400(3922)	450(4413)	550(5394)	650(6374)	850(8336)
770x770	820x820	870x870	980x980	1080x1080	1180x1180
1160x1090	1210x1140	1270x1190	1445x1365	1560x1480	1710x1650
700	750	800	900	1000	1200
1450	1550	1600	1850	2100	2400
300	350	350	400	450	500
750	800	800	950	1100	1200
5.7	5.7	10.0	14.6	14.6	20
210	210	220	220	230	230
23.7	25.7	29.4	36.4	42.7	50.8
19	24	28	33	50	65
8.2 x 1.9 x 2.2	8.8 x 2.0 x 2.2	9.3 x 2.1 x 2.2	9.8 x 2.5 x 2.2	10.4 x 2.5 x 2.4	11.8 x 2.7 x 2.5

SPECIFICATIONS



MODEL		SFV1800 (S.F.)	SFV2400 (S.F.)
Clamping Unit		IE2800	IE1360
Screw diameter	mm	70	60
Injection pressure	kg/cm ²	2083	1716
	Mpa	204	168
Injection holding pressure	kg/cm ²	1875	1544
	Mpa	184	151
Theoretical injection volume	cm ³	1347	763
Shot weight (PS)	g	1226	694
Injection rate (standard)	cm ³ /s	577	424
Injection rate (high speed)	cm ³ /s	-	848
Screw stroke	mm	350	270
Injection speed (standard)	mm/s	150	150
Injection speed (high speed)	mm/s	-	300
Plasticizing capacity (PS)	kg/h	321	260
Screw rotation speed	rpm	200	250
Clamping Unit			
Clamping force	ton(kN)	75 (735)	75 (735)
Daylight	mm	200	200
Rotary table size	Ø	1800	2400
Max. Daylight	mm	900	810
Min. Mold height	mm	700	610
Ejector force	ton(kN)	4.4 (42.8)	4.4 (42.8)
Ejector stroke	mm	80	95
Clamping Unit			
Motor capacity (standard)	kW	70.8	53.6
Motor capacity (high speed)	kW	-	89.0
Heater capacity	kW	27.6	26.9
Total electric power capacity (standard)	kW	98.4	80.5
Total electric power capacity (high speed)	kW	-	115.9
Hydraulic oil tank capacity	L	175	250
Machine weight	ton	22.5	17.5
Machine dimension (L×W×H)	m	7.8 x 2.9 x 2.8	8.2 x 2.9 x 2.8
Cooling water consumption	L/min	65	65

PLATEN DIMENSION

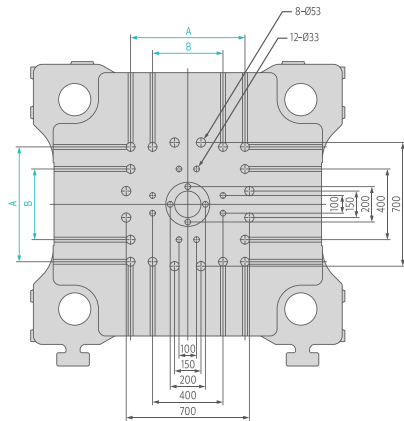
DL-A5 SUPER-FOAM

Unit: mm

DL500A5 (A: 500 / B: 250)

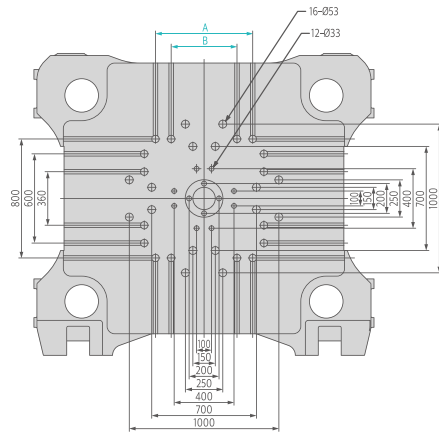
DL600A5 (A: 560 / B: 280)

DL700A5 (A: 650 / B: 400)

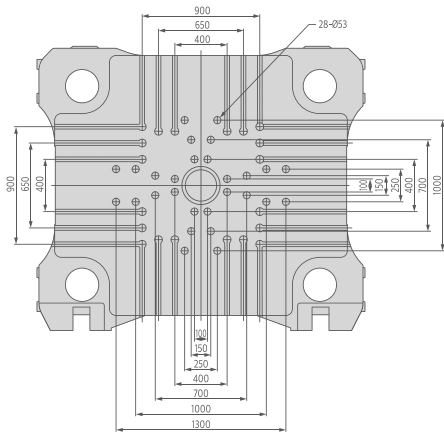


DL900A5 (A: 650 / B: 400)

DL1100A5 (A: 650 / B: 440)



DL1300A5



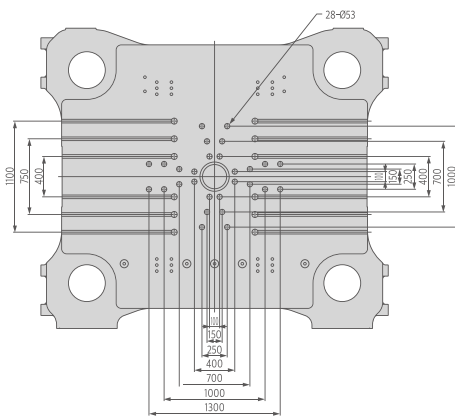
DL1800A5

DL2000A5

DL2300A5

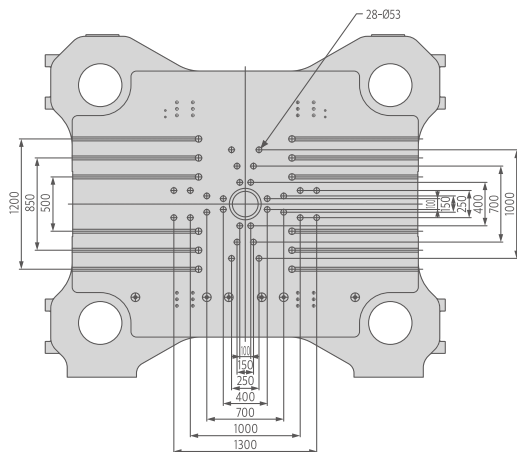
DL2500A5

DL2700A5



DL3000A5

DL3300A5

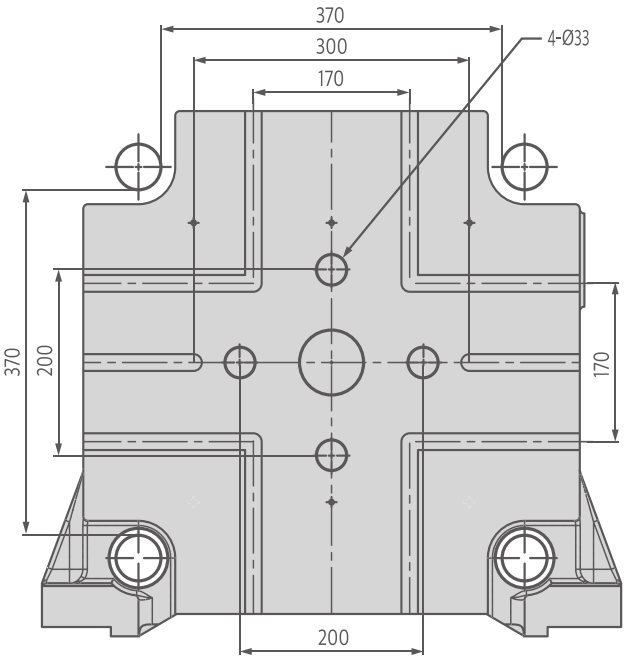


※ Product images may be subject to change without prior notice due to optional specifications or product improvements.

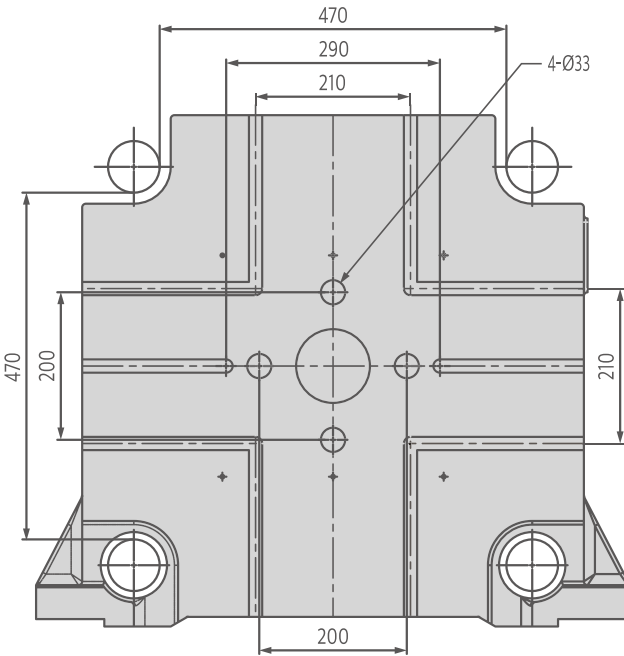
PLATEN DIMENSION

Unit: mm

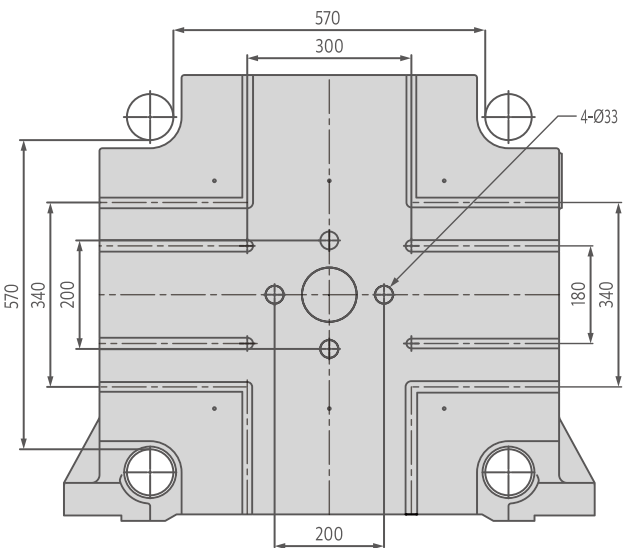
TE50A5



TE110A5



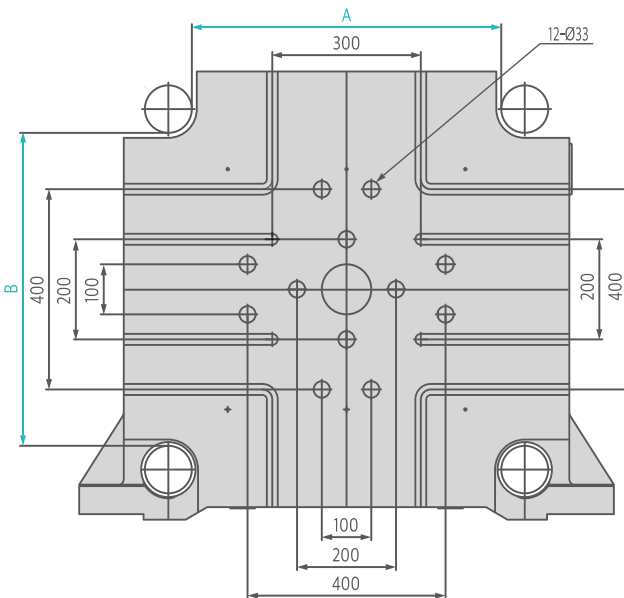
TE170A5



TE220A5 (A: 625 / B: 625)

TE280A5 (A: 670 / B: 670)

TE280WA5 (A: 720 / B: 720)



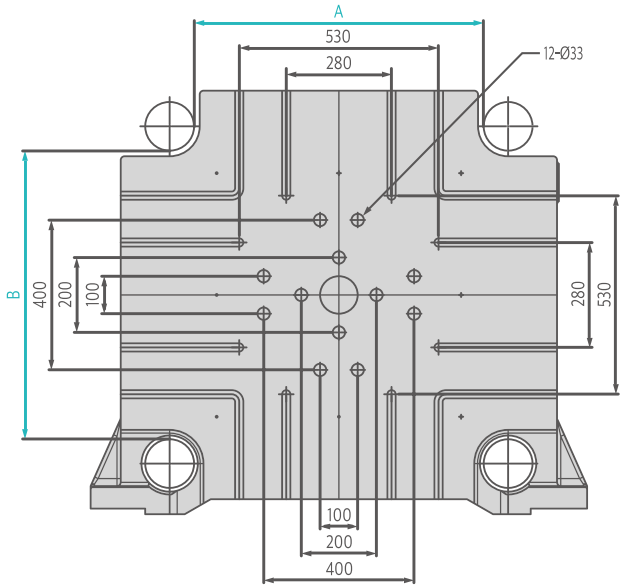
※ Product images may be subject to change without prior notice due to optional specifications or product improvements.

Unit: mm

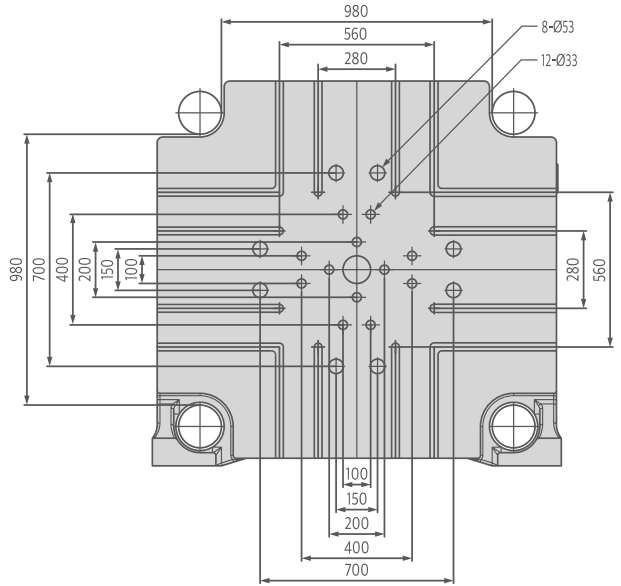
TE350A5 (A: 770 / B: 770)

TE400A5 (A: 820 / B: 820)

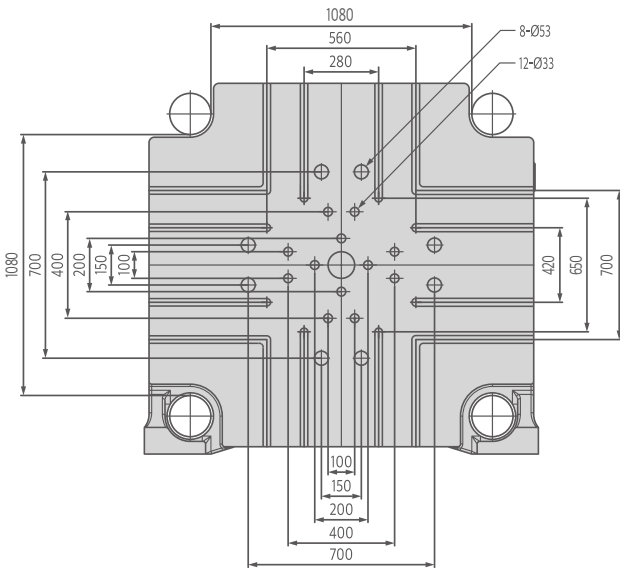
TE450A5 (A: 870 / B: 870)



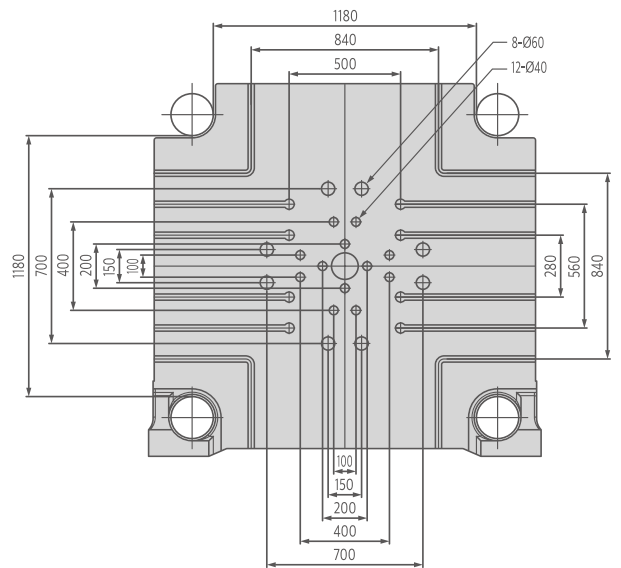
TE550A5



TE650A5



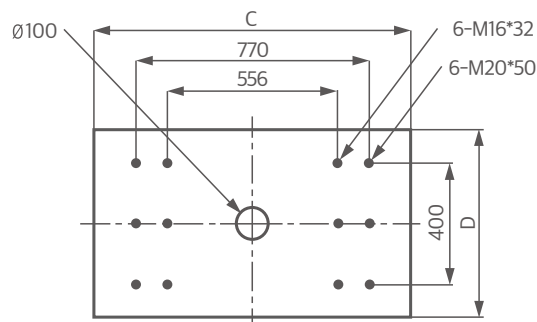
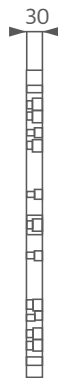
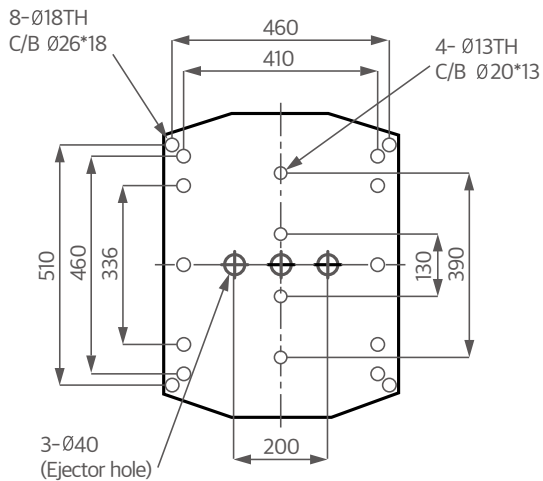
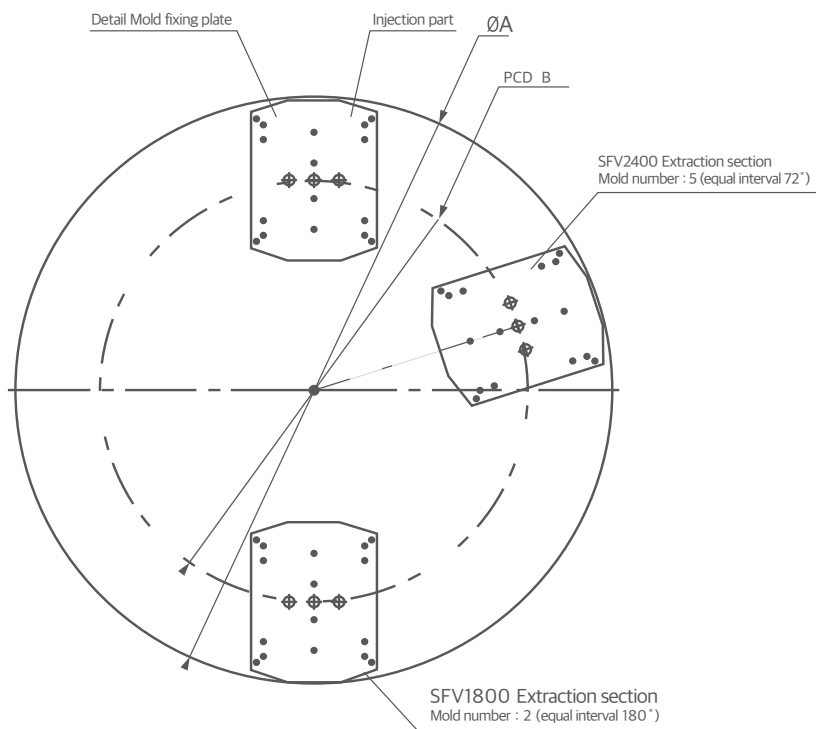
TE850A5



PLATEN DIMENSION



Unit: mm



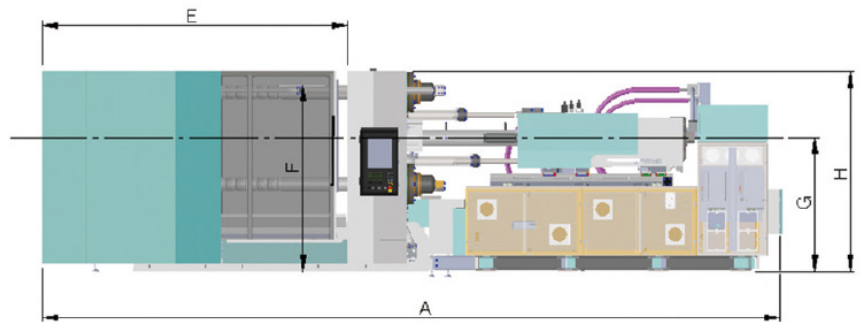
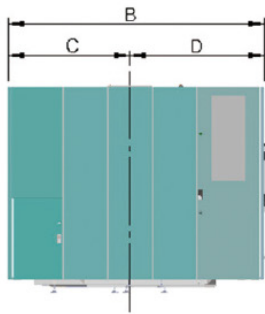
Turn-Table

Unit: mm	$\varnothing A$	B	C	D
SFV1800	1800	940	940	620
SFV2400	2400	1730	840	570

※ Product images may be subject to change without prior notice due to optional specifications or product improvements.

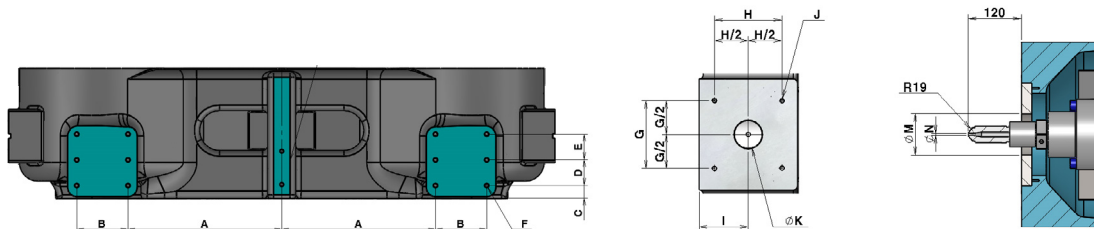
MACHINE DIMENSION

DL-A5 SUPER-FOAM



Machine Dimension

Unit: mm	A	B	C	D	E	F	G	H
DL500A5	8352	2713	1452	1261	3247	1970	1410	2144
DL600A5	8557	2939	1530	1409	3341	2025	1410	2202
DL700A5	9272	3052	1564	1488	3494	2130	1450	2350
DL900A5	10481	3319	1700	1619	4016	2245	1490	2461
DL1100A5	10685	3586	1820	1766	4347	2400	1600	2665
DL1300A5	12335	3815	1935	1880	5088	2586	1725	2858
DL1800A5	13768	4123	2109	2014	5716	3035	1969	3340
DL2000A5	14054	4423	2284	2139	6083	3065	1969	3364
DL2300A5	14054	4423	2284	2139	6083	3065	1969	3364
DL2500A5	15952	4623	2394	2229	6767	3344	2135	3688
DL2700A5	15952	4623	2394	2229	6767	3344	2135	3688
DL3000A5	17737	4973	2599	2374	6917	3607	2335	3977
DL3300A5	17737	4973	2599	2374	6917	3607	2335	3977



Robot installation position dimension

Hopper installation position dimension

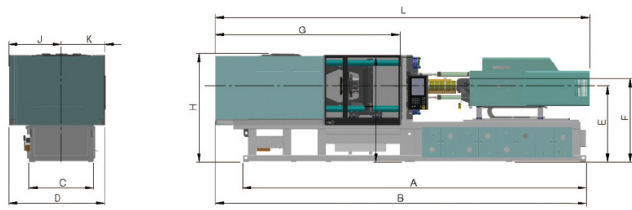
Nozzle dimension

Unit: mm	A	B	C	D	E	F	G	H	I	J	ØK	ØM	ØN
DL500A5	210	-	85	195	-	6-M20	165	165	120	4-M12	68	100	4
DL600A5	195	-	85	245	-	6-M20	165	165	130	4-M12	78	100	5
DL700A5	235	-	80	300	-	6-M24	165	165	140	4-M12	88	100	5
DL900A5	235	-	85	330	-	6-M24	165	165	170	4-M12	103	100	6
DL1100A5	335	-	85	375	-	6-M30	165	165	170	4-M12	103	100	6
DL1300A5	335	-	120	345	-	6-M30	280	200	190	4-M16	113	120	6
DL1800A5	495	-	115	485	-	6-M30	280	200	112	4-M16	123	120	7
DL2000A5	500	-	140	575	-	6-M36	280	200	112	4-M16	123	120	7
DL2300A5	500	-	140	575	-	6-M36	280	200	112	4-M16	123	120	7
DL2500A5	625	-	155	600	-	6-M36	280	200	132	4-M16	138	120	7
DL2700A5	625	-	155	600	-	6-M36	280	200	132	4-M16	138	120	7
DL3000A5	625	-	150	630	-	6-M36	280	200	218	4-M16	172	120	8
DL3300A5	625	-	150	630	-	6-M36	280	200	218	4-M16	172	120	8

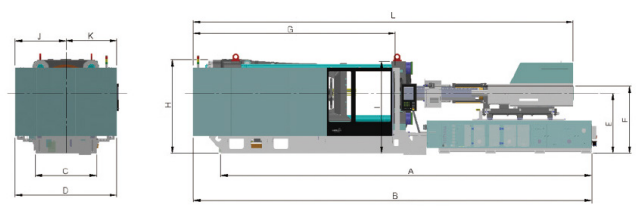
※ Product images may be subject to change without prior notice due to optional specifications or product improvements.

MACHINE DIMENSION

TE-A5 SUPER-FOAM



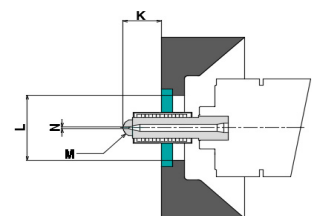
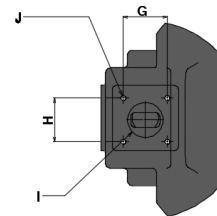
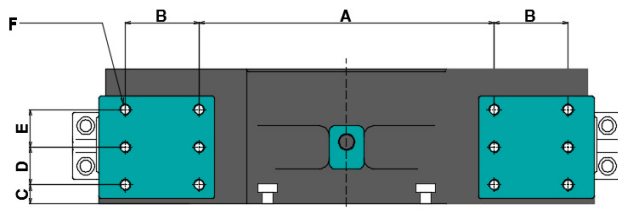
▲ 50Ton~400Ton



▲ 450Ton~850Ton

Machine Dimension

Unit: mm	A	B	C	D	E	F	G	H	I	J	K	L
TE50A5	3560	4067	980	1294	1250	1340	1839	1688	1535	647	647	4138
TE110A5	4320	4941	950	1294	1320	1420	2205	1786	1680	647	647	4846
TE170A5	4965	5676	1100	1574	1340	1455	2549	1913	1780	882	692	5687
TE220A5	5400	6017	1180	1644	1419	1534	2913	2038	1899	922	722	6157
TE280A5	5593	6267	1240	1713	1472	1602	3147	2038	1987	962	752	6827
TE280WA5	6398	7169	1240	1740	1497	1627	3290	2038	2042	963	777	7307
TE350A5	6735	7536	1410	1884	1565	1705	3595	2228	2165	1032	852	7977
TE400A5	7050	7741	1460	1949	1590	1730	3800	2217	2215	1072	877	8289
TE450A5	8226	8904	1410	2008	1505	1685	4061	2343	2145	1097	912	8727
TE550A5	8949	9747	1680	2469	1405	1595	4521	2173	2133	1235	1235	9818
TE650A5	9509	10357	1680	2495	1480	1715	4871	2320	2270	1248	1248	10498
TE850A5	10509	11397	1680	2765	1460	1695	5576	2450	2325	1383	1383	11503



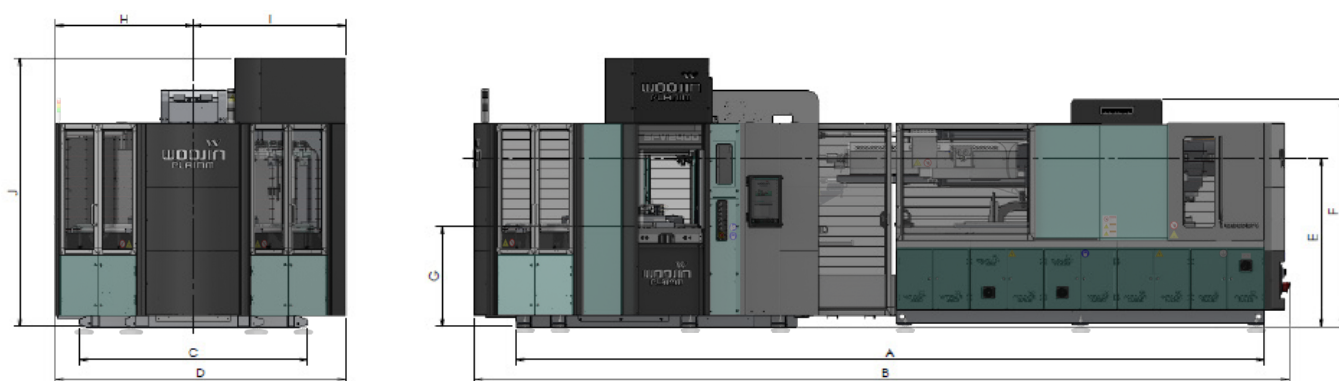
Robot installation position dimension

Hopper installation position dimension

Nozzle dimension

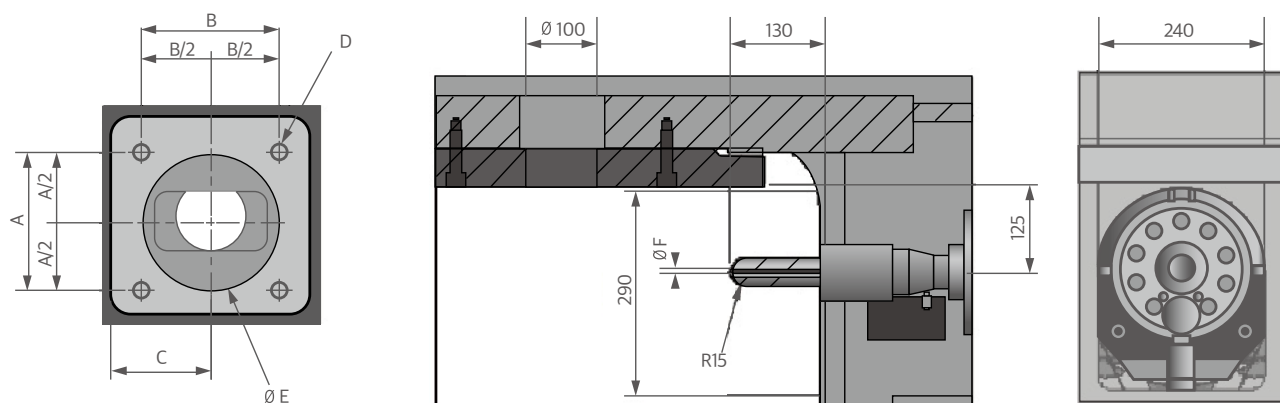
Unit: mm	A	B	C	D	E	F	G	H	ØI	J	K	ØL	M	ØN
TE50A5	340	85	30	75	-	8-M16 TAP DP32	80	80	70	4-M8 TAP	30	100	R9	2.5
TE110A5	455	85	35	70	-	8-M16 TAP DP32	100	100	80	4-M12 TAP	50	100	R9	2.5
TE170A5	420	140	35	70	70	12-M20 TAP DP40	100	100	80	4-M12 TAP	50	100	R9	3
TE220A5	560	140	35	70	70	12-M20 TAP DP40	100	100	80	4-M12 TAP	50	100	R9	3
TE280A5	560	140	35	140	-	8-M20 TAP DP40	120	120	100	4-M12 TAP	50	100	R14	3.5
TE280WA5	700	140	35	140	-	8-M20 TAP DP40	120	120	100	4-M12 TAP	50	100	R14	3.5
TE350A5	760	150	40	150	-	8-M20 TAP DP40	120	120	100	4-M12 TAP	50	100	R14	3.5
TE400A5	850	100	50	60	60	12-M24 TAP DP40	120	120	120	4-M12 TAP	50	100	R14	3.5
TE450A5	980	100	45	60	60	12-M24 TAP DP40	165	165	68	4-M12 TAP	130	100	R19	4
TE550A5	915	100	50	60	60	12-M24 TAP DP48	165	165	78	4-M12 TAP	130	100	R19	5
TE650A5	1000	240	70	105	105	12-M24 TAP DP48	165	165	88	4-M12 TAP	130	100	R19	5
TE850A5	1120	240	70	105	105	12-M24 TAP DP48	165	165	103	4-M12 TAP	130	100	R19	6

※ Product images may be subject to change without prior notice due to optional specifications or product improvements.



Machine Dimension

Unit: mm	A	B	C	D	E	F	G	H	I	J
SFV1800	7450	7764	2560	2843	1800	2672	1025	1521	1322	2543
SFV2400	7469	8123	2210	2899	1710	2295	1025	1380	1520	2703



▲ Hopper installation position dimension

▲ Nozzle dimension

Hopper installation position dimension

Nozzle dimension

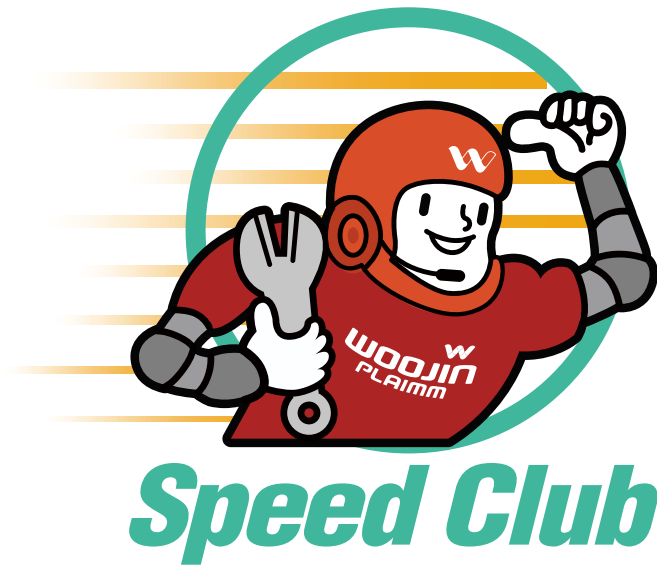
Unit: mm	A	B	C	D	E	F
SFV1800	165	165	100	4-M12	68	8
SFV2400	120	120	85	4-M12	120	8

※ Product images may be subject to change without prior notice due to optional specifications or product improvements.

OPTION LIST

Injection Unit	Clamping Unit	General
Standard		
01. Automatic Injection unit swiveling (Below IH 11900)	01. Safety Foot-board (Above 900ton)	01. Standard Maintenance tools
02. Single Flight Screw	02. Automatic safety Door open/close (Above 500ton)	02. Standard spare part
03. Injection valve gate circuit (AC 1 + DC 1)	03. Clamping area Curtain sensor	03. Leveling pad
04. Back-Pressure Closed-loop system	04. Hydraulic Core puller (Moving platen side, 1 stage)	04. Cooling water distributor
05. PID Heating Control	05. Air blow-off unit (Fixed side 1 + Moving side 1)	05. Automatic grease lubrication (Clamping)
06. Weekly Heating Timer	06. Safety device(for electric & hydraulic)	06. Robot interface (Standard)
07. Cold screw start protection mode	07. Spring mold mode	07. 3 Phase electric outlet (2 ea)
08. Temperature display & Alarm in abnormal Temp.	08. Automatic Mold thickness adjust mode	08. Single Phase electric outlet (1 ea)
09. Auto Purging	09. Mold-Open Speed & Pressure step (5 step)	09. Steel tray for resin leakage
10. Injection Speed & Pressure step (10 step)	10. Ejector Speed & Pressure step (3 step)	10. Hopper throat temperature control device
11. Holding Speed & Pressure step (5 step)		11. Hydraulic oil purification device
12. Charging Speed & Pressure step (3 step)		12. Hydraulic oil temperature control device
13. Back Pressure control step (3 step)		13. Hydraulic oil level alarm
14. Injection Pressure Graph Display		14. Hydraulic oil temperature check & alarm
15. Injection Speed Graph Display		15. Hydraulic oil heating mode
16. Screw RPM Display		16. 3 color alarm light
17. Cushion Display & Alarm		17. Shot data saving by external way
18. Charging time count & alarm		18. Production data statistics
19. Screw & Barrel (Anti Wear & Corrosive)		19. Alarming & History save
20. Measuring electricity		20. Log history save
21. Raw material supply device		21. I/O circuit display
22. Shut-off Nozzle (Pneumatic)		22. Shot data save (Internal 1,000 / External device)
Option		
01. Heater Disconnection check device	01. Rotating Core Circuit	01. Hydraulic Auto-Clamp unit
02. Hopper Slide (L/M)	02. Safety Foot-board (Below 700ton)	02. Anchor-bolt set (Clamping unit)
03. Hopper Ladder & Stand	03. Core & Ejector on Fly	03. Heater Insulation Band
04. Valve Gate Circuit & Connector (Interior type)	04. Daylight Extension	04. Automatic Grease Lubrication (Injection unit)
05. Hydraulic Valve Gate Block (Interior type)	05. Core-Back Mode	05. Robot Interface (EM12, EM 67, EM67.1, SPI)
06. Pneumatic Valve Gate Block (Interior type)	06. Core Pressure release Circuit (Automatic)	06. CMS (Central Monitoring System)
07. Hydraulic Valve Gate Device (External device)	07. Core Pressure release Circuit (Manual)	07. AVR (Automatic Voltage Regulator) on Electric Panel
08. Nozzle cylinders equipped with Potentiometers	08. Product Chute	08. UPS (Uninterruptable Power Supply) on Electric Panel
09. Charging on Fly (AC Motor)	09. Automatic Tie-bar Retraction	09. Dosing unit Interface (for Masterbatch)
	10. Hydraulic Core Check Valve	10. Gas Injection Interface
	11. Hydraulic Core Interlock Connector (EM13, WJ Standard)	11. Steam Injection Interface
	12. Hydraulic Core Puller (2~8 Stages)	12. External Temperature Display (F/P)
	13. Mold ring on Moving-Platen	13. Interior type Hot Runner Controller (EM13, WJ Standard)
	14. Spring type Ejector retraction	14. Booster
	15. Ejector Check Valve	15. Nitrogen generator
	16. Ejector Interlock Connector (WJ Standard, EM13)	16. Air dryer
	17. Ejector Forward / Backward External switch	17. C.P.M system (Counter Pressure Molding System)
	18. Mold Insulation Plate	
	19. Pneumatic Core Puller (1-7 Stages)	

Injection Unit	Clamping Unit	General
Standard		
01. Single Flight Screw 02. Injection valve gate circuit (AC 1 + DC 1) 03. Back-Pressure Closed-loop system 04. PID Heating Control 05. Weekly Heating Timer 06. Cold screw protection mode 07. Temperature display & Alarm in abnormal Temp. 08. Auto Purging 09. Injection Speed & Pressure step (10 step) 10. Holding Speed & Pressure step (5 step) 11. Charging Speed & Pressure step (3 step) 12. Back Pressure control step (3 step) 13. Injection Pressure Graph Display 14. Injection Speed Graph Display 15. Screw RPM Display 16. Cushion Display & Alarm 17. Charging time count & alarm 18. Screw & Barrel (Anti Wear & Corrosive) 19. Raw material supply device 20. Shut-off Nozzle (Pneumatic)	01. Ejecting on fly 02. Air blow-off unit (Fixed side 1 + Moving side 1) 03. Safety device(for electric & hydraulic) 04. Automatic clamp force measurement mode 05. Automatic Mold thickness adjust mode 06. Mold-Open Speed & Pressure step (5 step) 07. Mold-Close Speed & Pressure step (5 step) 08. Ejector Speed & Pressure step (3 step)	01. Standard Maintenance tools 02. Standard spare part 03. Leveling pad 04. Cooling water distribution 05. Automatic Ball-screw grease lubrication (All parts) 06. Automatic oil lubrication (Toggle) 07. Robot interface (Standard) 08. 3 Phase electric outlet (2 ea) 09. Single Phase electric outlet (1 ea) 10. Hopper throat temperature control device 11. 3 color alarm light 12. Shot data saving by external way 13. Production data statistics 14. Alarming & History save 15. Log history save 16. I/O circuit display 17. Shot data save (Internal 1,000 / External device)
Option		
01. Heater Disconnection check device 02. Hopper Slide (L/M) 03. Long-holding pressure type upgrade 04. Valve Gate Circuit & Connector (Interior type) 05. Pneumatic Valve Gate Block (Interior type) 06. Nozzle cylinders equipped with Potentiometers	01. Rotating Core Circuit 02. Safety Foot-board (Above 650ton) 03. Core on Fly 04. Daylight Extension 05. Automatic safety Door open/close 06. Product Chute 07. Hydraulic Core Interlock Connector (EM13, WJ Standard) 08. Hydraulic Core Device (Fixed : 170~400ton / Moving / 1or2 stage) 09. Mold ring on Moving-Platen 10. Ejector Interlock Connector (WJ Standard, EM13) 11. Ejector Forward/Backward External switch 12. Mold Insulation Platen 13. Pneumatic Core Puller (1-3 Stages)	01. Lubricating oil Recycling device 02. Product drop check device 03. Product quality sorting device (Below 280ton) 04. Hydraulic Auto-Clamp unit 05. Steel tray for resin leakage 06. Heater Insulation Band 07. Robot Interface (EM12, EM67, EM67.1, SPI) 08. CMS (Central Monitoring System) 09. AVR (Automatic Voltage Regulator) on Electric Panel 10. UPS (Uninterruptable Power Supply) on Electric Panel 11. Dosing unit Interface (for Masterbatch) 12. Gas Injection Interface 13. Steam Injection Interface 14. External Temperature Display (F/P) 15. Interior type Hot Runner Controller (EM13, WJ Standard) 16. Booster 17. Nitrogen generator 18. Air dryer 19. C.P.M system (Counter Pressure Molding System)



- 1 24-Hour On-Site Support
- 2 Dedicated B/S Team
- 3 Mobile App Support
- 4 Extended Parts Warranty

1 Expert Technical Staff

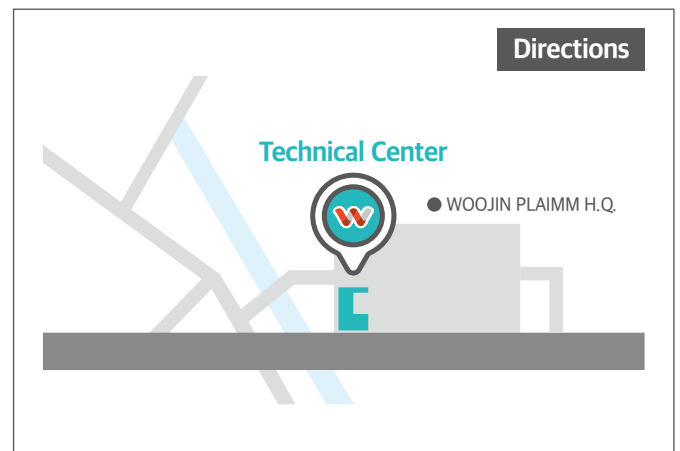
| Over 10 Years of Experience in Injection Molding (Optimization / Foaming / Trial Injection)

2 Technical Research Team

| Control technology and mechanical design personnel available for ongoing support

3 Auxiliary Facility Support

| Deployment of Skilled Technicians and Consulting Staff



WOOJIN PLAIMM Technical Center **delivers the best results** to our customers through **strategic consulting** and **technical knowledge sharing**.



GLOBAL NETWORK



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- ※ Content updated as of January 2, 2026.

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